

Ollama Installation and Maintenance Guide

(C) Sujeet Banerjee

Overview

This guide installs Ollama and downloads the models used in the bootcamp: `llama3.2` (LLM) and `nomic-embed-text` (embedding model).

System Requirements

1. Windows 10/11 or macOS 13+
2. 8 GB RAM minimum (16 GB recommended)
3. 10 GB free disk space
4. Internet connection (to download models)
5. ~~Docker Desktop is NOT required~~
6. ~~Python is NOT required for Day 1~~ (*Required for Day2/Day3*)

Windows 10/11

1. Visit <https://ollama.com/download/windows> and download the Windows installer.
2. Run the installer.
3. Open PowerShell or Command Prompt.
4. Verify installation:

```
ollama --version
```

If successful, download the models:

```
ollama pull llama3.2
ollama pull nomic-embed-text
```

Run the LLM:

```
ollama run llama3.2
```

The Ollama server normally listens on `http://localhost:11434`.

macOS

1. Visit <https://ollama.com/download/mac> or install via Homebrew:

```
brew install ollama
```

2. Start Ollama (if needed):

```
ollama serve
```

3. Verify installation:

```
ollama --version
```

Download the models:

```
ollama pull llama3.2
ollama pull nomic-embed-text
```

Run the LLM:

```
ollama run llama3.2
```

Useful Commands

```
ollama list          # List downloaded models
ollama rm llama3.2   # Remove a model
ollama show llama3.2 # Model information
ollama serve         # Start server manually
```

Change LLM Hyperparameters

From within Ollama CLI or Terminal:

```
/?                List available LLM commands
/set parameter ... Set a (hyper)parameter
/set verbose       Show LLM stats
/set quiet         Disable LLM stats
/set think <high|medium|low> Enable thinking
/set nothink       Disable thinking
/set parameter num_ctx <int> Set the context size
/set parameter temperature <float> Set creativity level
```

Troubleshooting

- Do not use Office-Laptop (certain ports might be blocked due to office compliance)
- If 'ollama' is not recognized, restart the terminal after installation.
- Firewall software may block `localhost:11434`.
- Verify the server by opening `http://localhost:11434` in a browser; a simple response indicates the service is running.

Ollama Maintenance: Cleaning Up Local Logs, Memory, and Storage

To ensure Ollama runs smoothly without consuming unnecessary system resources, you can periodically clean up stale data, memory, and storage.

1. Freeing Up Memory (RAM/VRAM)

When a model is running, it stays loaded in memory for a period of time (typically 5 minutes by default). To immediately unload the model and free up RAM/VRAM, run the `ollama stop` command:

```
ollama stop llama3.2
```

Alternatively, you can restart the Ollama service or app to clear all loaded models from memory.

2. Cleaning Up Model Storage (Disk Space)

Downloaded models can take up significant disk space. To remove a model you no longer need:

```
ollama rm <model_name>
```

Example: `ollama rm llama3.2`

Manual Cache Deletion: If you want to clear the entire model cache (or if `ollama rm` fails), you can delete the models directory directly:

- **Windows:** Delete the contents of `C:\Users\<YourUsername>\.ollama\models`
- **macOS:** Delete the contents of `~/.ollama/models`

3. Clearing Local Logs

Ollama generates logs that can grow over time.

- **macOS:** The log files are usually located at `~/.ollama/logs/`. You can delete old log files in this directory to free up space.
- **Windows:** Logs are typically found in `%LOCALAPPDATA%\Ollama\logs` or the `.ollama` directory.